TASK-3

Select and Explain Three SSL Methods for the Dataset

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Outline

Choice of SSL

SSL1: SimCLR

SSL2: DINOv2

SSL3: BYOL

ChoiceofSSL

SSL Method	Key Mechanism	Best For	Rice Dataset Advantage
SimCLR	Contrastive (Pos/Neg)	Invariance	Ignores camera angle/lighting variations
BYOL	Reconstructive (Masking)	Texture Depth	Learns the microscopic “fabric” of the grain
DINOv2	Distillation (Teacher/Student)	Shape/Semantics	Automatically finds grain boundaries and edges

WhatisSSL?

- Learns without labels
- Finds patterns automatically
- Great for image datasets

SimCLRVisualizatio

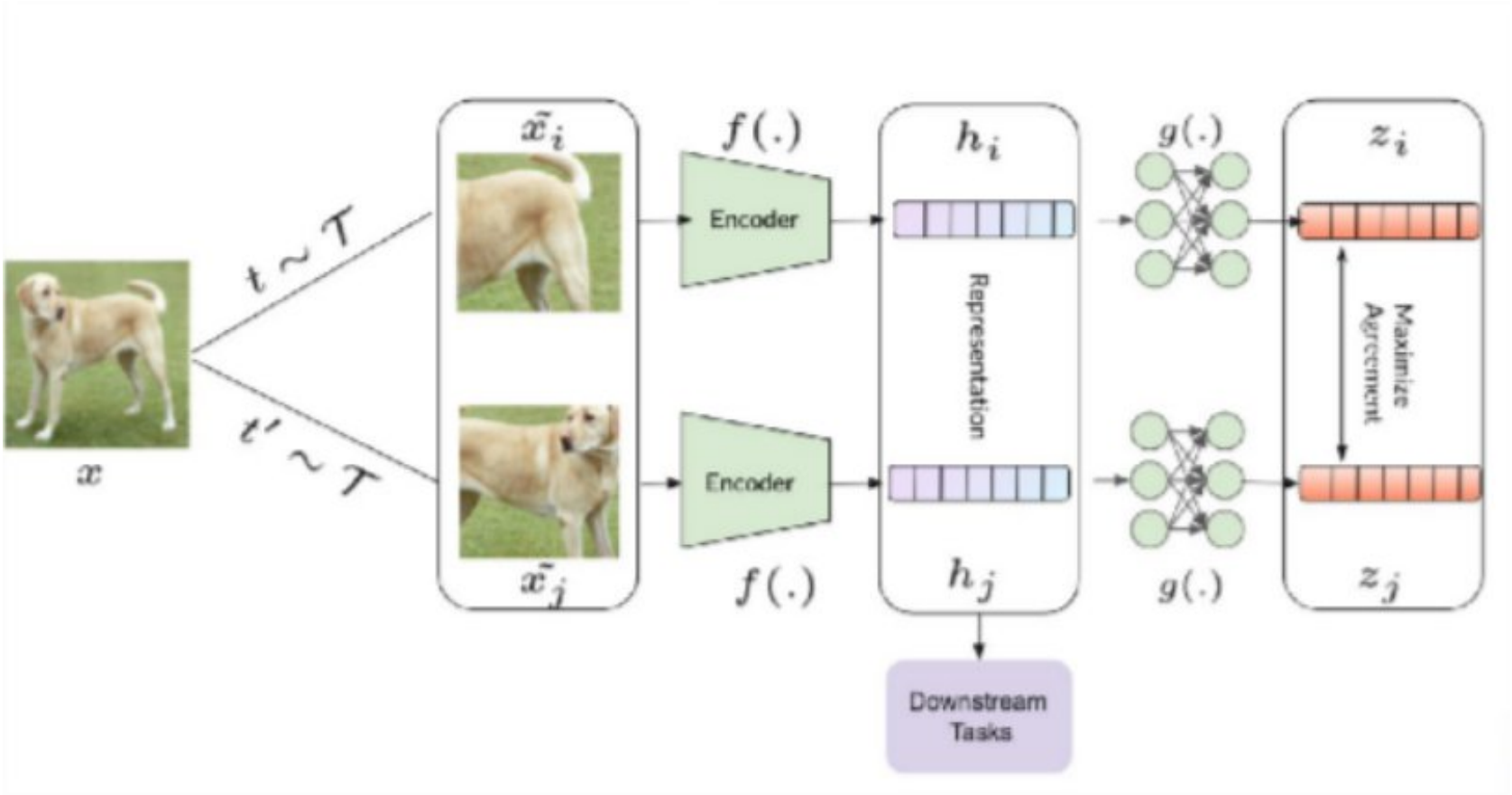

Figure 1: SimCLR Visualization

SimCLR(SimpleFrameworkforContrastiveLearning)

Key Idea: Learn representations by maximizing agreement between augmented views.
Pipeline:

- Input image $\rightarrow$ augment twice (x_i, x_j)
- Encoder $f(\cdot)$ (ResNet) $\rightarrow$ projection head $g(\cdot)$
- Get embeddings z_i, z_j

Contrastive Loss (NT-Xent):

$$L_{i,j} = -\log \Sigma \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)}$$

Numerical Example:

- $\mathrm{sim}(z_i, z_j) = 0.8$, others ≈ 0.2, $\tau = 0.5$
- $\exp(1.6) = 4.95$, others $\exp(0.4) = 1.49$

Loss Calculation:

$$L \approx -\log \frac{4.95}{4.95 + 3 \times 1.49} \approx 0.74$$

Key Components:

- Strong augmentations
- Large batch size
- Projection head improves representation

DINOv2Visualization

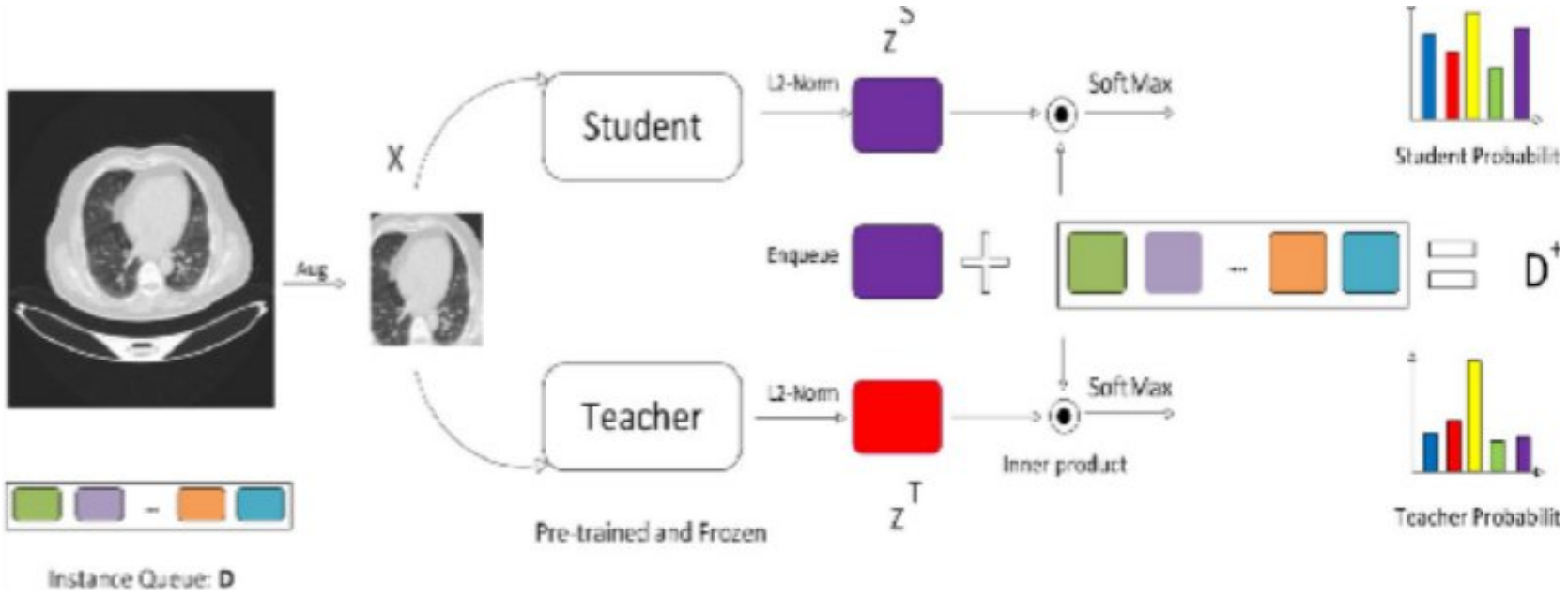

Figure 2: DINOv2 Visualization

DINOv3(Self-DistillationwithNoLabels)

Key Idea: Student-teacher self-distillation with centering + sharpening.

Architecture:

- Student network (learns)
- Teacher network (EMA of student)

Method:

- Multiple views (global + local crops)
- Teacher produces soft targets
- Student matches distributions

Loss (Cross-Entropy):

$$L = -\sum p_t(x) \log p_s(x)$$

Numerical Example:

- Teacher: $[0.7, 0.3]$, Student: $[0.6, 0.4]$

Loss Calculation:

$$-(0.7 \log 0.6 + 0.3 \log 0.4) \approx 0.61$$

Key Components:

- Centering (avoid collapse)
- Sharpening (low entropy targets)
- No negative pairs needed

BYOLVisualization

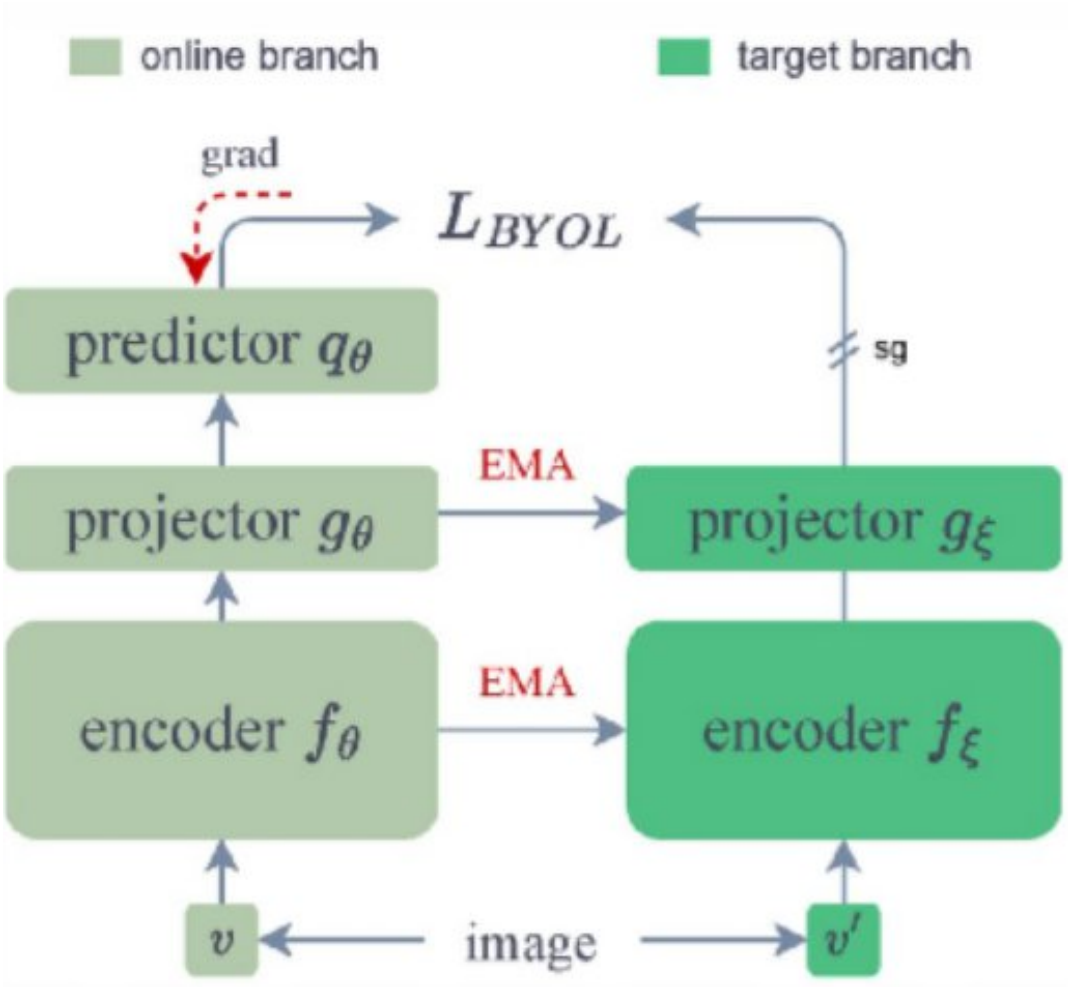

Figure 3: BYOL Visualization

BYOL(BootstrapYourOwnLatent)

Key Idea: No negative pairs; uses online and target networks.

Architecture:

- Online network: f_θ, projector, predictor
- Target network: f_ξ (EMA of θ)

Algorithm:

- Two views x, x'
- Online predicts target representation
- Update target via EMA

Loss Function:

$$L = \left\| q_\theta(z_\theta) - \mathrm{sg}(z_\xi) \right\|_2^2$$

Numerical Example:

- $q_\theta = [0.6, 0.8]$, $z_\xi = [0.5, 0.7]$

Loss Calculation:

$$(0.6 - 0.5)^2 + (0.8 - 0.7)^2 = 0.01 + 0.01 = 0.02$$

Key Components:

- No contrastive loss
- Stop-gradient (sg)
- EMA target stabilizes training